MILITARY STANDARD

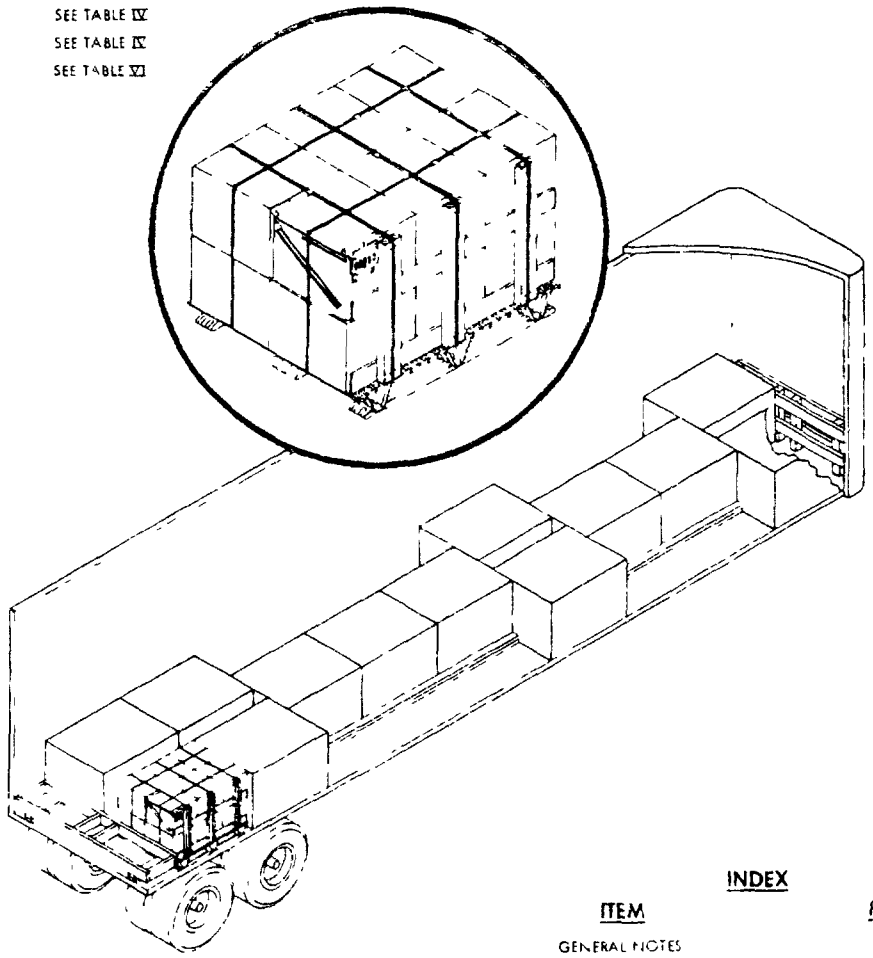
MIL-STD-1320-1/8
(NAVY)

2 NOVEMBER 1976

TRUCKLOADING
CARTRIDGE, 20MM, M50 SERIES (LOOSE)
IN CONTAINER M548
DOMESTIC AND FLEET ISSUE LOADS

UNIT LOAD DATA

UNIT LOAD DWG	WR 53/870 & WR-54/294
DIMENSIONS	SEE TABLE IV
CUBE	SEE TABLE IV
WEIGHT	SEE TABLE IV
DOT HAZARD CLASS	SEE TABLE VI



INDEX

ITEM	PAGE
GENERAL NOTES	1
FTL & LT	2
LOAD PATTERNS	3
TYPICAL LOADS	5 & 6

NOTES

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS REFER TO INDEX OF STANDARDS, MIL HDBK 220 (NAVY)

FSC 8140

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GENERAL NOTES

1. THIS DOCUMENT PROVIDES DETAILED INSTRUCTIONS FOR TRUCKING FLEET ISSUE AND DOMESTIC UNIT LOADS OF THE SERIES M50 20MM CARTRIDGES (LOOSE) IN THE CONTAINER M548 ALL UNIT LOADS SHALL BE IN ACCORDANCE WITH WR-53/870 AND WR-54/294
2. THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40, 42, 44 AND 45 FT TRAILERS WITH BOTH NAILABLE AND NON-NAILABLE FLOORS
3. LENGTH OF THE STRUTS PIECE 5 IN DETAIL B SHALL BE SUFFICIENT TO CAUSE THE BLOCKING ASSEMBLY TO FILL THE VOID BETWEEN THE LADING AND THE TRAILER DOORS WHEN THE DOORS ARE CLOSED

WARNING

DO NOT USE TRAILERS WITH ROLL UP DOORS

4. WHEN THE TRAILER HAS NON-NAILABLE FLOORS AND SLEEPERS CANNOT BE NAILED TO THE FLOOR TO PREVENT LATERAL MOTION SUBSTITUTE THE SPACER ASSEMBLY, DETAIL C PAGE 4 FOR THE SLEEPERS
5. THE SPACER ASSEMBLY SHALL FILL THE VOID BETWEEN THE UNIT LOAD AND TRAILER SIDE WALLS OR BETWEEN UNIT LOADS A SLIP FIT MAY BE USED PROVIDED THE CUMULATIVE SLACK BETWEEN UNIT LOAD (S) TRAILER WALL AND SPACER ASSEMBLY DOES NOT EXCEED ONE INCH
6. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS
7. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS TO INSIDE OF TRAILER IN AN ACCESSIBLE AREA CLOSE AND SEAL TRAILER DOORS ATTACH APPROPRIATE PLACARD FOR TYPE OF 20MM CARTRIDGES BEING SHIPPED TO BOTH SIDES FRONT AND BACK OF TRAILER (SEE TABLE VI PAGE 4)
8. APPLICABLE MATERIAL SPECIFICATIONS DUNNAGE LUMBER MM-L-751 NAILS FF N 101 TYPE II STYLE 10 COMMON BRIGHT STRAPPING QQ-S-781 TYPE I CLASS A
9. FOR GENERAL TRUCK LOADING PROCEDURES REFER TO THE GENERAL TRUCK LOADING DOCUMENT MIL-STD-1320 (NAVY)

FTL AND LTL 40, 42, 44 & 45 FT TRAILERS (VAN)

WARNING

THIS STANDARD PROVIDES FOR INCREASED LOADINGS WHICH PRODUCE HEAVIER CROSS TANDEM AXLE WEIGHTS AND GROSS VEHICLE WEIGHTS THESE INCREASED WEIGHTS ARE NOT AUTHORIZED IN ALL 50 STATES BEFORE LOADING MORE THAN 12 UNIT LOADS IN A TRAILER DETERMINE THAT THE LIMITATIONS OF THE NOTES IN TABLES I II & III ARE MET

1. TABLES I II AND III PROVIDE THE CORRECT LOAD PATTERNS FOR THE LENGTH OF THE TRAILER BEING LOADED AND THE ALLOWABLE GROSS TANDEM AXLE WEIGHT AND GROSS VEHICLE WEIGHTS
2. BOTH LIGHT WEIGHT AND STANDARD WEIGHT TRACTOR-TRAILERS MAY BE USED FOR ALL LOAD PATTERNS ONLY THE GROSS TANDEM AXLE WEIGHT AND GROSS VEHICLE WEIGHT RESTRICTIONS MUST BE MET

NOTE

LIGHT WEIGHT TRACTOR-TRAILER - TARE WEIGHT OF TRAILER TANDEM AXLES 6 800 LBS MAXIMUM AND THE DRIVE TANDEM AXLES 10 500 LBS MAXIMUM
STANDARD WEIGHT TRACTOR-TRAILER TARE WEIGHT OF TRAILER TANDEM AXLES 7 800 LBS MAXIMUM AND THE DRIVE TANDEM AXLES 11 500 LBS MAXIMUM

THE ABOVE MAXIMUM WEIGHTS MAY BE ADJUSTED WHEN EXPERIENCE INDICATES THAT THE WEIGHT RESTRICTIONS CAN BE MET

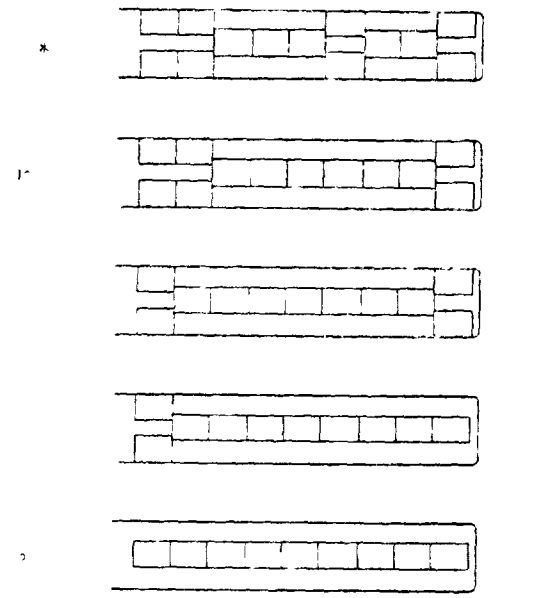
2. TO DETERMINE THE CORRECT LOAD PATTERN TO USE SELECT THE CORRECT TABLE FOR THE LENGTH OF TRAILER BEING LOADED OPPOSITE THE NUMBER OF UNIT LOADS TO BE LOADED FIND THE CORRECT LOAD PATTERN MAKE CERTAIN THAT THE QUANTITY BEING SHIPPED IS COMPATIBLE WITH THE WEIGHT LAWS OF THE STATES THROUGH WHICH THE SHIPMENT IS ROUTED BLOCK AND BRACE USING THE PRINCIPLES SHOWN ON PAGES 5 & 6
3. THE TRUCKLOAD SHOWN ON PAGE 5 DOES NOT LIMIT SHIPMENT TO THE NUMBER OF UNIT LOADS SHOWN IT IS PRESENTED TO SHOW THE BASIC PRINCIPLES OF BLOCKING ONLY THE LOAD PATTERN TABLES SHOW THE CORRECT NUMBER OF UNIT LOADS FOR THE LENGTH OF THE TRAILER BEING LOADED AND THE WEIGHT RESTRICTIONS OF THE ROUTING
4. WHEN SHIPPING LESS THAN THE AMOUNTS SHOWN IN THE LOAD TABLES FOLLOW THE PROCEDURES SHOWN ON PAGE 6 DO NOT USE TRAILERS WITH NON-NAILABLE FLOORS

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TABLE I

40 FT TRAILER

NO. OF UNIT LOADS | LOAD PATTERN | FWD →

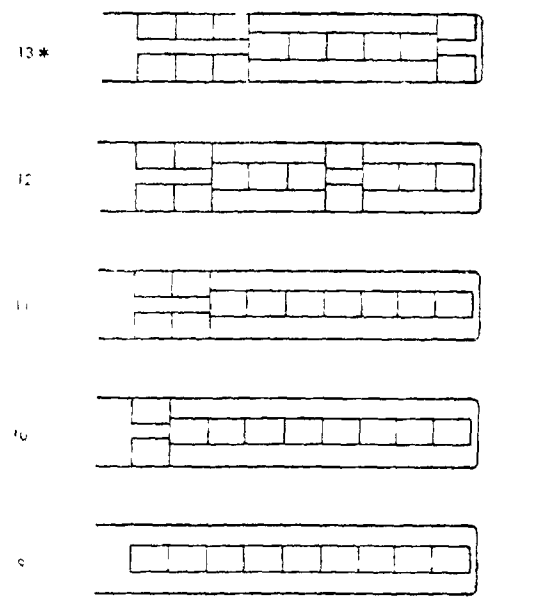


OR LESS CONTINUE SINGLE ROW PATTERN

TABLE II

42 FT TRAILER

NO. OF UNIT LOADS | LOAD PATTERN | FWD →

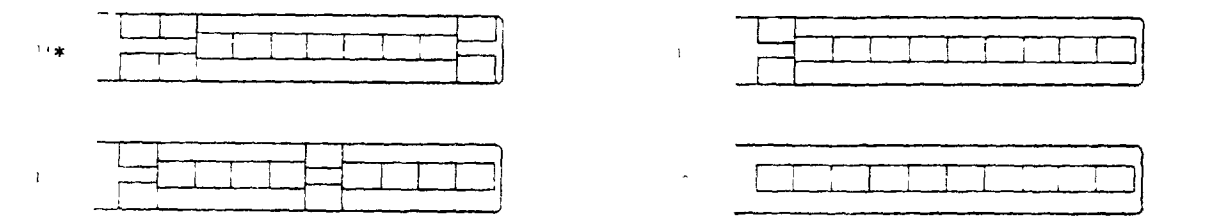


OR LESS CONTINUE SINGLE ROW PATTERN

TABLE III

44 & 45 FT TRAILERS

NO. OF UNIT LOADS | LOAD PATTERN | FWD →



9 OR LESS CONTINUE SINGLE ROW PATTERN

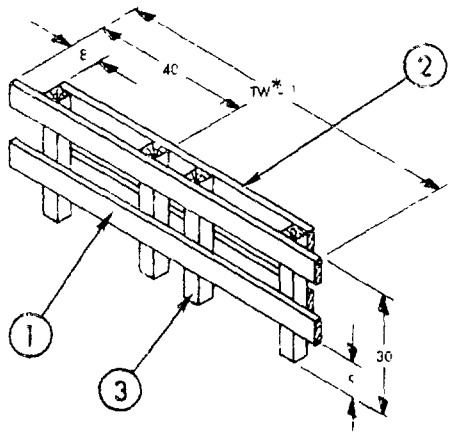
* AUTHORIZED WHEN ROUTING IS THROUGH STATES PERMITTING 34 000 LBS OR MORE ON THE TANDEM AXLES AND A GROSS VEHICLE WEIGHT OF 80 000 LBS. BOTH STANDARD AND LIGHT WEIGHT TRACTOR TRAILERS MAY BE USED.

NOTE: WHEN SHIPPING 12 OR LESS UNIT LOADS, BOTH STANDARD AND LIGHT WEIGHT TRACTOR-TRAILERS MAY BE USED IN ALL 50 STATES WITHOUT EXCEEDING THE WEIGHT LAWS.

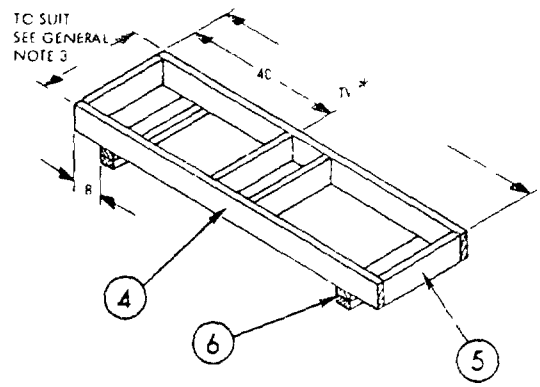
NOTE: ALL LOAD PATTERNS HAVE THE UNIT LOADS POSITIONED IN THE TRAILER WITH THE 48 1/4 INCH DIMENSION OF THE UNIT LOAD CROSSWISE TO THE TRAILER.

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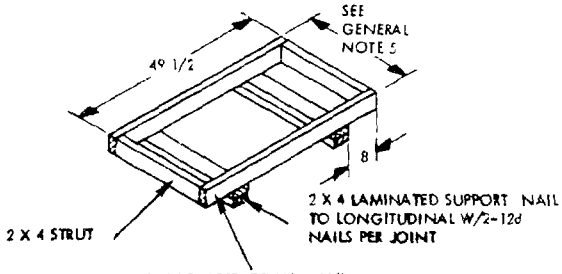
* Tw = TRAILER WIDTH



DETAIL A
FRONT BULKHEAD
1 REQUIRED



DETAIL B
REAR BLOCKING ASSEMBLY
1 REQUIRED



DETAIL C
SPACER ASSEMBLY
(FOR NON-NAILABLE FLOORS)

TABLE IV

UNIT LOAD DATA

DWC NO	DIMENSIONS	CUBE (CU FT)	WEIGHT (LBS) APPROX
WR-54/294	49 5/8 L X 40 1/4 W X 34 H	39.3	3580
WR-53/870	49 5/8 L X 40 1/4 W X 34 3/4 H	40.2	3582

TABLE V

TRUCKLOAD DATA

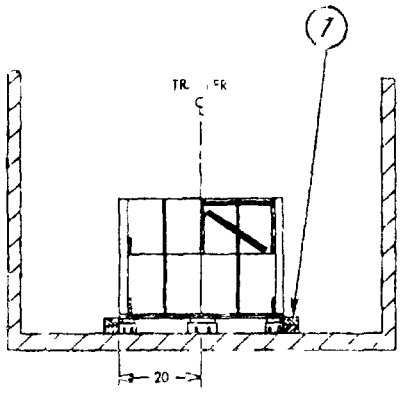
NO OF UNIT LOADS	WEIGHT (LBS) APPROX
13	46,540
12	42 960
11	39,380
10	35,800
9	32,220

TABLE VI

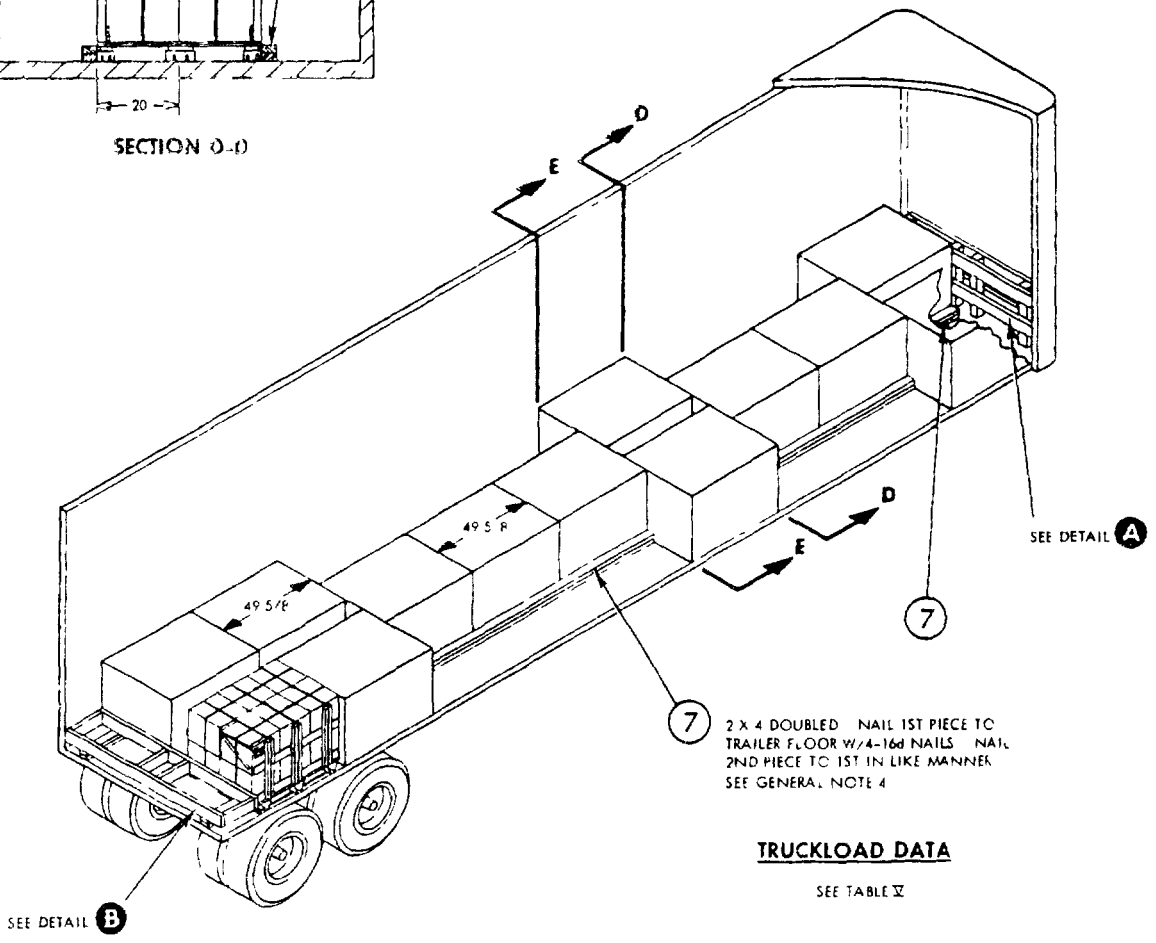
DOT HAZARD CLASSIFICATION

DODIC / NALC NC	CARTRIDGE	HAZARD CLASS
A651	M222 TP-T	C
A655	M242 HEI-T	A
A890	M56 HEI	A
A891	M55 TP	C
A892	M54 HTP	C
A921	M51 DUMMY	INERT

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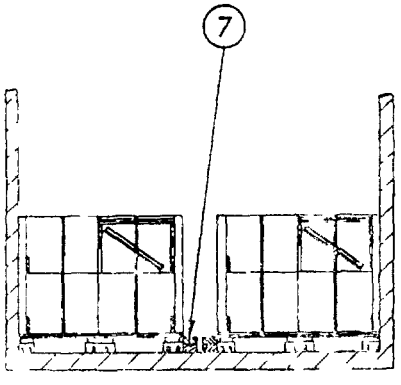
SECTION D-D



2 X 4 DOUBLED NAIL 1ST PIECE TO TRAILER FLOOR W/4-16d NAILS NAIL 2ND PIECE TO 1ST IN LIKE MANNER SEE GENERAL NOTE 4

TRUCKLOAD DATA

SEE TABLE V



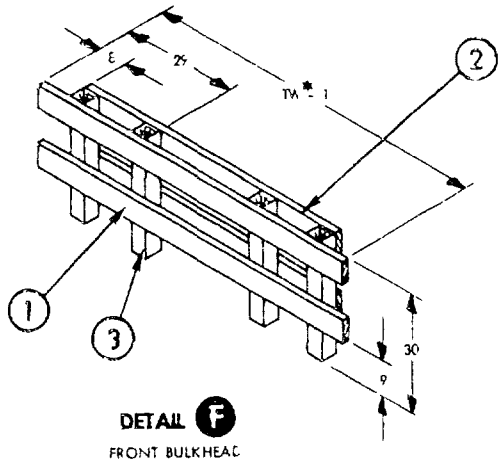
SECTION E-E

*TW - TRAILER WIDTH

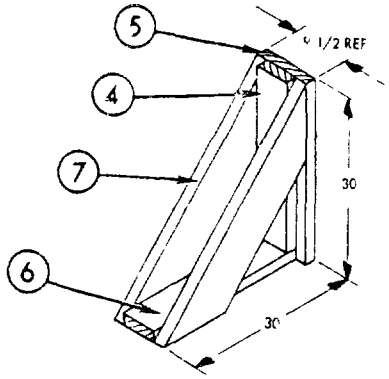
PIECE NO	DESCRIPTION	SIZE	NO OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
7	SLEEPER	2 X 4 X 48	36	SEE FIELD NOTE		
6	SUPPORT	2 X 4 TO SUIT	4	4 6	2/JOINT 3	10d 10d
5	STRUT	2 X 6 TO SUIT	4	SEE 4		-
4	CROSSMEMBER	2 X 6 X TW* - 1	2	5	3/JOINT	16d
3	VERTICAL	4 X 4 X 30	4	SEE 1 & 2	-	-
2	FWD CROSSMEMBER	2 X 6 TO SUIT	2	3	3/JOINT	12d
1	AFT CROSSMEMBER	2 X 6 X TW* - 1	2	3	3/JOINT	12d

LIST OF MATERIALS & NAILING DATA

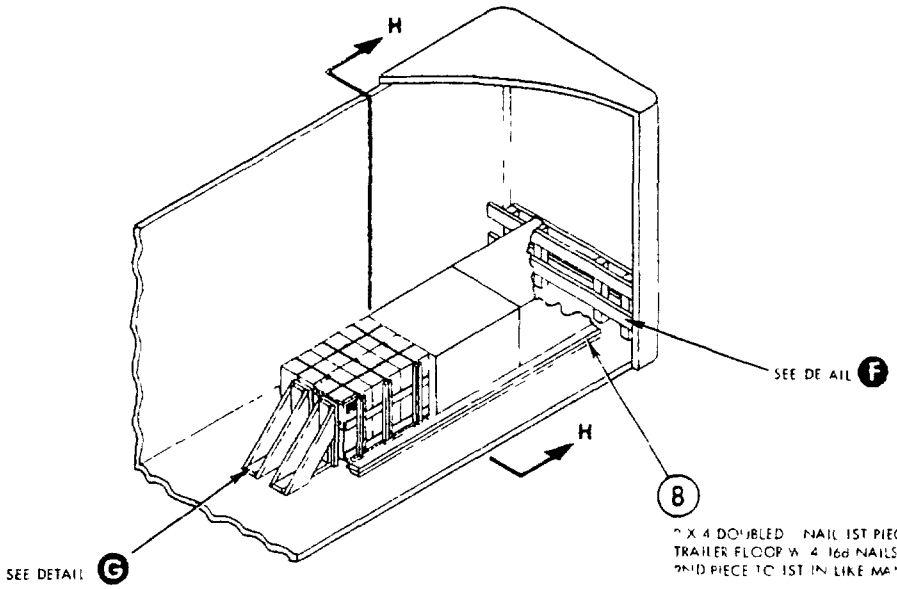
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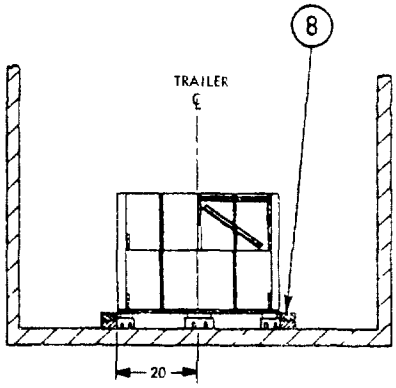
DETAIL F
FRONT BULKHEAD
1 REQUIRED



DETAIL G
KNEE BRACE
2 REQUIRED



2 X 4 DOUBLED NAIL 1ST PIECE TO TRAILER FLOOR W/ 4 16d NAILS NAIL 2ND PIECE TO 1ST IN LIKE MANNER



SECTION H-H

* T TRAILER WIDTH

PIECE NO	DESCRIPTION	SIZE	NO OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
	LIST OF MATERIALS & NAILING DATA				NAILS	
8	SLEEPER	2 X 4 X 48		SEE FIELD NOTE		
7	BRACE DIAGONAL	2 X 6 TO SUIT	4	1 & 6	4 PER JOINT	16d
6	BRACE CLEAT	2 X 6 X 28 1/2	2	TRAILER FLOOR	6	10d
5	BRACE BACKUP	2 X 10 X 30	2	6	6	16d
4	BRACE UPRIGHT	2 X 6 X 29 1/2	2	5	4	8d
3	VERTICAL	4 X 4 X 30	4	SEE 1 & 2		
2	FWD CROSSMEMBER	2 X 6 TO SUIT	1	3	3 PER JOINT	10d
1	AFT CROSSMEMBER	2 X 6 X T*		3	3 PER JOINT	20d

TYPICAL SINGLE ROW LTL

PREPARING ACTIVITY
NAVY - OS

REVIEW ACTIVITY
NAVY - OS
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